

Q.1	Q.2	Q.3	Q.4	Q.5	TOTAL
-----	-----	-----	-----	-----	-------

Q.1) (15) Data comes in various formats, impacting how it can be stored, analyzed, and utilized. Distinguish between the following data types by providing real-world examples for each:

Structured DataDefinition:

Structured data is highly organized information that follows a predefined format, typically stored in rows and columns of a relational database. It's characterized by a fixed schema, making it easy to search, sort, and analyze.

Clear and relevant example:

A customer database in a store. This database would have columns for customer ID, name, address, purchase history, etc. Each row represents an individual customer with their specific information.

Unstructured DataDefinition:

Unstructured data lacks a predefined format and organization. It exists in its natural state and doesn't fit neatly into rows and columns.

Clear and relevant example:

An email you receive from a friend. The email contains text, attachments like images or documents, and may not follow a specific structure throughout.

Semi-structured DataDefinition:

Semi-structured data falls somewhere between structured and unstructured. It has some internal organization but doesn't adhere to a strict schema like a relational database. It often uses tags or markers to define elements within the data.

Clear and relevant example:

An XML file containing product information. The data might have elements like <product>, <name>, <price>, and <description>, providing some organization without the rigidity of a database table.

KEYS TO FINAL EXAM

(This is a 100-minute exam)

Q.2) (25) A telecommunications company like Turk Telekom is experiencing a high customer churn rate (i.e., customers cancelling their service). To develop strategies for customer retention, they aim to predict which customers are at risk of churning. The company has collected customer data, including demographics, service plans, usage patterns, and past support interactions. Here is a sample dataset in CSV format based on the problem definition:

CUSTID	Gender	Age	ContractType	MonthlyCharges	TotalCharges	Service Usage	Support Calls	Churn
1	Female	45	One year	70.35	1200.50	High	2	No
2	Male	32	Month-to-month	56.75	450.20	Medium	5	Yes
3	Female	29	Two year	89.50	2400.75	Low	1	No
4	Male	52	One year	45.00	1100.00	Medium	3	No
5	Female	37	Month-to-month	99.99	150.99	High	7	Yes
6	Male	41	Two year	60.80	1800.40	Low	0	No
7	Female	34	One year	75.20	900.00	Medium	4	No
8	Male	25	Month-to-month	40.50	300.00	High	6	Yes
9	Female	55	Two year	55.00	1700.00	Low	1	No
10	Male	38	One year	85.75	1300.50	Medium	2	No
11	Female	28	Month-to-month	65.30	600.75	High	8	Yes
12	Male	46	Two year	70.00	2000.00	Low	0	No

Column Descriptions:**CUSTID:** Unique identifier for each customer.**Gender:** Gender of the customer.**Age:** Age of the customer.**ContractType:** Type of contract (Month-to-month, One year, Two year).**MonthlyCharges:** The amount charged to the customer monthly.**TotalCharges:** The total amount charged to the customer.**ServiceUsage:** Overall usage level of the service (Low, Medium, High).**SupportCalls:** Number of support calls made by the customer.**Churn:** Indicates whether the customer has churned (Yes, No).

Churn, in the context of business and telecommunications, refers to the phenomenon where customers stop using a company's products or services and leave for a competitor or discontinue the service altogether. This is often measured as a churn rate, which is the percentage of customers who cancel or do not renew their subscriptions over a given period. In the dataset provided, the Churn column indicates whether a customer has churned or not:

- Yes: The customer has canceled their service and is no longer using the company's products.
- No: The customer has not canceled their service and is still using the company's products.

KEYS TO FINAL EXAM

(This is a 100-minute exam)

a) (10) Identify a suitable supervised learning algorithm for predicting customer churn in this scenario. Briefly explain why this algorithm is appropriate for this problem.

An appropriate supervised learning algorithm for predicting churn in this scenario is a random forest classifier. Predicting customer churn is a classification problem. We want to classify customers into two categories: churned (Yes) or not churned (No) based on their characteristics. Random Forest excels at classification tasks. The data set contains numeric data (age, monthly charges, etc.) and categorical data (gender, contract type, etc.). Random Forest can effectively handle both types of data. Random Forest is less prone to overfitting the training data compared to some other algorithms. This is important because we want the model to generalize well to unseen customer data. While Random Forest models aren't perfectly interpretable, they do offer some level of explanation compared to complex algorithms like neural networks. This allows us to understand which features are most important in predicting churn.

KEYS TO FINAL EXAM

(This is a 100-minute exam)

b) (10) Describe the key steps involved in building a customer churn prediction model using the chosen algorithm. Briefly explain what happens at each step.

- Check for missing data in any features.
- Convert categorical data (e.g., Gender, ContractType) into numerical values suitable for the model.
- Standardize numerical features (e.g., Age, MonthlyCharges) to a similar scale to prevent features with larger ranges from dominating the model.
- Divide the dataset into two parts: training data (used to train the model) and testing data (used to evaluate the model's performance on unseen data). A common split is 80% for training and 20% for testing.
- Train the Random Forest Classifier on the training data. This involves creating multiple decision trees, each making predictions based on a random subset of features and data points.
- Adjust hyperparameters (e.g., number of trees in the forest) to potentially improve the model's performance.
- Evaluate the model's performance on the testing data using metrics like accuracy, precision, recall, and F1-score.
- Analyze the importance of each feature in the model's predictions. This helps understand which customer attributes are most influential in predicting churn.
- Once satisfied with the model's performance, deploy it to score new customer data and identify those at risk of churning.

c) (5) Once you have a trained model, how can the company use it to identify customers at risk of churning?

With a trained customer churn prediction model, Turk Telekom can identify customers at risk of churning through a scoring process:

- New customer data, containing features like the training data (e.g., demographics, service usage), is collected. This could be real-time data or batch data fed into the model periodically.
- Similar to the training process, any new data goes through pre-processing steps like handling missing values, encoding categorical data, and scaling numerical features to ensure compatibility with the model.
- The prepped customer data is fed into the trained Random Forest model. Each customer receives a churn probability score, typically ranging from 0 (low churn risk) to 1 (high churn risk).
- A churn risk threshold is set based on the model's performance and business needs. For example, Turk Telekom might choose a threshold of 0.7, meaning customers with a predicted churn probability above 0.7 are considered high risk.
- Customers are segmented based on their churn risk score. High-risk customers (scores above the threshold) are prioritized for intervention.

Based on customer segmentation and potentially other factors, Turk Telekom can develop targeted retention strategies:

- Personalized Offers: Providing high-risk customers with exclusive discounts, bundled services, or loyalty programs to incentivize them to stay.
- Proactive Support: Reaching out to high-risk customers to address any service issues or concerns they might have.

KEYS TO FINAL EXAM

(This is a 100-minute exam)

Q.3) (20) A credit card company is constantly on the lookout for fraudulent transactions. They receive a massive volume of transaction data daily, including details like purchase amount, location, merchant category, and time of purchase from physical POS (Point of Sale) devices. Physical POS is a payment device allowing a company to receive payments from all debit and credit cards. The company wants to develop a system to identify potentially fraudulent transactions in real time.

Here is a sample dataset in CSV format based on the problem definition. This dataset includes various features that could help in identifying fraudulent transactions, such as purchase amount, location, merchant category, and time of purchase:

TransactionID,CardholderID,PurchaseAmount,Location,MerchantCategory,TimeOfPurchase,Fraudulent

```
1,1001,50.25,New York,Groceries,2024-05-29 08:15:30,No
2,1002,4500.00,California,Electronics,2024-05-29 14:22:10,Yes
3,1003,75.50,Montana,Restaurants,2024-05-29 12:35:50,No
4,1004,200.75,New York,Clothing,2024-05-29 09:45:20,No
5,1005,5.00,Florida,Convenience Store,2024-05-29 07:05:15,No
6,1006,4500.00,Alabama,Jewelry,2024-05-29 16:50:45,Yes
7,1007,120.00,Illinois,Gas Stations,2024-05-29 11:30:05,No
8,1008,35.40,New York,Entertainment,2024-05-29 20:15:25,No
9,1002,3000.00,Nevada,Luxury Goods,2024-05-29 12:55:55,Yes
10,1010,60.00,Boston,Groceries,2024-05-29 13:20:35,No
11,1011,85.75,Washington,Restaurants,2024-05-29 10:40:00,No
12,1012,40.50,Texas,Pharmacy,2024-05-29 08:25:10,No
13,1006,5000.00,Washington,Electronics,2024-05-29 16:45:30,Yes
14,1014,15.75,Oregon,Convenience Store,2024-05-29 06:45:20,No
15,1015,80.25,Minnesota,Clothing,2024-05-29 14:30:00,No
16,1019,225.00,New York,Jewelry,2024-05-29 17:05:10,Yes
17,1017,95.60,Nevada,Entertainment,2024-05-29 12:50:45,No
18,1018,60.00,San Francisco,Gas Stations,2024-05-29 09:15:25,No
19,1019,3500.00,Arizona,Luxury Goods,2024-05-29 17:06:55,Yes
20,1020,25.00,Rhode Island,Groceries,2024-05-29 11:20:35,No
21,1021,45.75,Indiana,Pharmacy,2024-05-29 08:30:10,No
22,1022,150.00,Wyoming,Electronics,2024-05-29 20:50:30,No
23,1023,10.50,New York,Convenience Store,2024-05-29 07:45:20,No
24,1018,4000.00,Hawai,Jewelry,2024-05-29 18:30:00,Yes
25,1025,30.75,Colorado,Restaurants,2024-05-29 13:35:10,No
```

Column Descriptions:

TransactionID: Unique identifier for each transaction.

CardholderID: Unique identifier for each cardholder.

PurchaseAmount: Amount of money spent in the transaction.

Location: Location where the transaction occurred.

MerchantCategory: Category of the merchant (e.g., Groceries, Electronics, Restaurants).

TimeOfPurchase: Timestamp of when the purchase was made (in the format YYYY-MM-DD HH:MM).

Fraudulent: Indicates whether the transaction was identified as fraudulent (Yes, No).



KEYS TO FINAL EXAM

(This is a 100-minute exam)

Describe one anomaly detection technique you could employ to analyze credit card transaction data. Briefly explain the concept behind the technique.

An effective anomaly detection technique for analyzing credit card transaction data is the Isolation Forest algorithm. Isolation Forest is an ensemble-based anomaly detection method specifically designed to handle outliers or anomalies in a dataset. The fundamental idea behind the Isolation Forest algorithm is that anomalies are "few and different" compared to normal instances. Thus, anomalies can be isolated more easily.

Random Partitioning:

The algorithm creates multiple trees (forest) by recursively partitioning the data.

Each tree is constructed by randomly selecting a feature and then randomly selecting a split value between the minimum and maximum values of the selected feature.

Isolation:

In this process, anomalies, which are few in number, tend to be isolated in fewer splits compared to normal instances.

The path length from the root of the tree to the terminating node (where the instance gets isolated) is used as an indication of normality. Shorter paths correspond to anomalies.

Anomaly Score:

The algorithm calculates an anomaly score based on the average path length of an instance across all trees.

The score determines the likelihood of the instance being an anomaly. A high score indicates an anomaly, while a low score indicates a normal instance.

KEYS TO FINAL EXAM

(This is a 100-minute exam)

Q.4) (20) A streaming service offers a vast library of movies and TV shows. They have collected user data on viewing habits, including genres watched, duration of viewing sessions, and time of day for watching. The streaming service wants to understand their customer base better and personalize their recommendations.

Here is a sample dataset in CSV format based on the problem definition. This dataset includes user data on viewing habits, such as genres watched, duration of viewing sessions, and time of day for watching:

UserID	Age	Gender	GenresWatched	Avg SessionDuration	Total WatchTime	Preferred WatchTime
1	25	Male	Action Comedy	45	1200	Night
2	34	Female	Drama Romance	60	1500	Evening
3	29	Male	Documentary Science	30	800	Morning
4	42	Female	Action Thriller	50	2000	Night
5	23	Male	Comedy Drama	35	900	Afternoon
6	37	Female	Horror Mystery	40	1000	Evening
7	31	Male	Action Sci-Fi	55	1800	Night
8	28	Female	Romance Comedy	45	1100	Afternoon
9	45	Male	Drama Historical	60	2200	Evening
10	30	Female	Sci-Fi Fantasy	35	950	Morning
11	26	Male	Comedy Action	40	1050	Night
12	33	Female	Mystery Thriller	50	1250	Evening

Column Descriptions:

UserID: Unique identifier for each user.

Age: Age of the user.

Gender: Gender of the user.

GenresWatched: Genres of content the user frequently watches (multiple genres separated by '|').

AvgSessionDuration: Average duration of each viewing session in minutes.

TotalWatchTime: Total watch time in minutes.

PreferredWatchTime: Preferred time of day for watching content (Morning, Afternoon, Evening, Night).

KEYS TO FINAL EXAM

(This is a 100-minute exam)

a) (10) Identify a suitable unsupervised learning algorithm for segmenting the users based on their viewing habits. Briefly explain why this approach is appropriate for this problem.

A suitable unsupervised learning algorithm for the segmentation of users based on their viewing habits in this scenario is the K-means clustering algorithm.

We don't have predefined categories for user segments. K-means helps discover inherent groups (clusters) based on user behavior (viewing habits) in the data. K-means is a popular technique for customer segmentation tasks. It can group users with similar viewing habits (genres, session duration, watch time) into distinct clusters. K-means allows some level of interpretation after clustering. By analyzing the centroids (representative data points) of each cluster, we can understand the characteristics of each user segment. While other unsupervised algorithms like Hierarchical clustering could also be explored, K-means offers a good balance of effectiveness, interpretability, and scalability for this user segmentation problem.

b) (10) Describe the key steps involved in segmenting the users using the chosen algorithm. Briefly explain what happens at each step.

To segment users based on their viewing habits using K-means clustering:

Handling Missing values: Check for missing data and potentially impute missing values or remove rows/columns with excessive missing entries.

Feature Engineering: Consider creating new features that might be informative for clustering. For instance, you could create a feature representing the total number of genres a user watch frequently.

Feature Scaling: Standardize numerical features (e.g., AvgSessionDuration, TotalWatchTime) to ensure all features contribute equally to the clustering process.

Number of Clusters (k) Selection: Determine the optimal number of user segments (k). This can be done using techniques like the Elbow Method, which involves evaluating the sum of squared distances within each cluster for different k values. The "elbow" on the plot suggests the optimal number of clusters.

K-means Initialization: Randomly select k data points as initial centroids, representing the center of each cluster.

Iterative Clustering:

a. Calculate the distance between each user and all centroids. Assign each user to the cluster with the nearest centroid.

b. Recalculate the centroid (representative data point) for each cluster based on the users now assigned to it.

Repeat steps a and b until the centroids no longer significantly change their positions indicating convergence.

While K-means doesn't provide a formal evaluation metric, you can assess the compactness (how tightly grouped users are within a cluster) and separation (how distinct the clusters are from each other) to validate the clustering results. Analyze the centroids and user assignments to understand the characteristics of each segment. For instance, one segment might represent users who prefer watching short action comedies at night, while another might be users who enjoy longer documentaries in the morning.

KEYS TO FINAL EXAM

(This is a 100-minute exam)

Q.5) (20) A retail store has provided you with a sample of their daily sales data for the past week in a CSV file named "sales_data.csv". The data includes columns such as date, product category (category), product ID (product_id), quantity sold (quantity), and total sales amount (sales_amount).

Sample Data (sales_data.csv):

```
date,category,product_id,quantity,sales_amount
2024-05-20,Electronics,TV123,2,1200.00
2024-05-20,Clothing,Shirt456,1,35.99
2024-05-21,Appliances,Fridge789,1,899.99
2024-05-21,Grocery,Milk123,3,5.99
2024-05-22,Electronics,Monitor007,1,250.00
2024-05-22,Clothing,Pants987,2,79.98
2024-05-23,Appliances,Washer345,2,599.98
2024-05-23,Grocery,Eggs567,1,2.99
```

a) (10) Calculate the total sales per product category for the past week

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")
sales_df["date"] = pd.to_datetime(sales_df["date"])
today = pd.to_datetime('today')
last_week_start = today - pd.Timedelta(days=7)
print(sales_df[ sales_df["date"] >= last_week_start ].groupby( ["category"] )[ "sales_amount" ].sum())
```

b) (10) Identify the top 5 products (by product ID) that generated the most revenue during the past week.

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")
sales_df["date"] = pd.to_datetime(sales_df["date"])
today = pd.to_datetime('today')
last_week_start = today - pd.Timedelta(days=7)
total_revenue_per_product = sales_df.groupby('product_id')['sales_amount'].sum()

sorted_revenue = total_revenue_per_product.sort_values(ascending=False)

top_5_products = sorted_revenue.head(5)
print(top_5_products)
```